



quickservice – LOCALIZED SERVICE DISCOVERY & BOOKING APP

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- Thank you

Introduction

Have you ever needed a nearby service provider but didn't know whom to contact? Quick Serve alias Service spot is planned to make this easier by helping people quickly find and book local services in one place. The goal is to create a simple and practical solution to a common everyday problem.



Problem Statement

- In daily life, people struggle to find reliable nearby service providers quickly. Traditional search methods (local shops, word-of-mouth) are slow and lack transparency in pricing and availability. Service providers also lack a digital presence to reach customers efficiently.
- In many cases, users are forced to depend on a **single service provider**, leading to **single vendor lock-in**, which reduces flexibility and limits choice.
- There is a need for a platform that connects users with verified local service providers in real time.

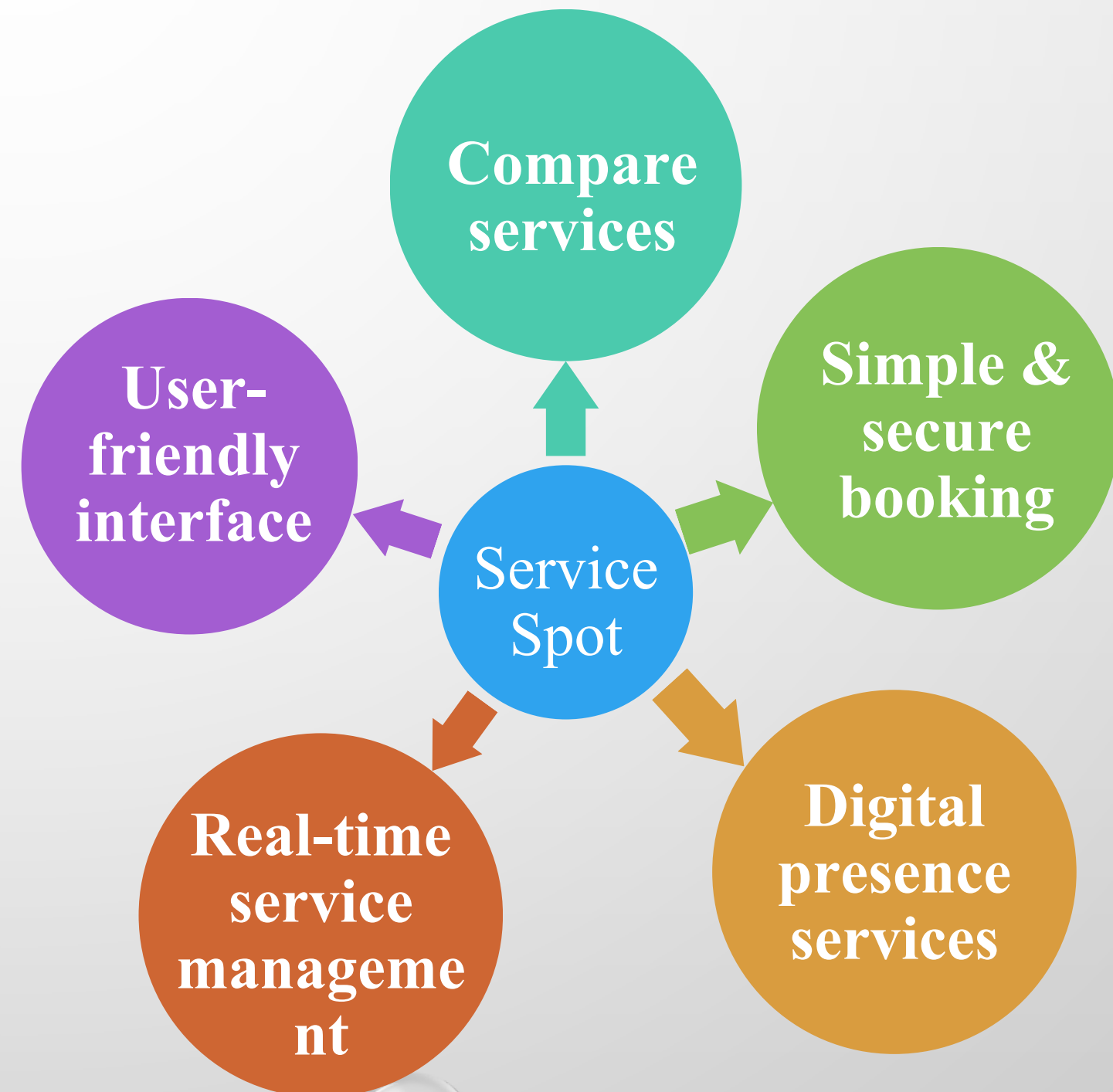


What is QuickServe?

A convenient app for discovering local services easily using map.

Objectives

- Provide a **user-friendly interface** to search local services
- Help users **compare** providers based on rating, cost, and distance
- Offer **simple & secure booking** system
- Build a **digital presence** for small/local service businesses
- Enable **real-time service management** (status, history, notifications)

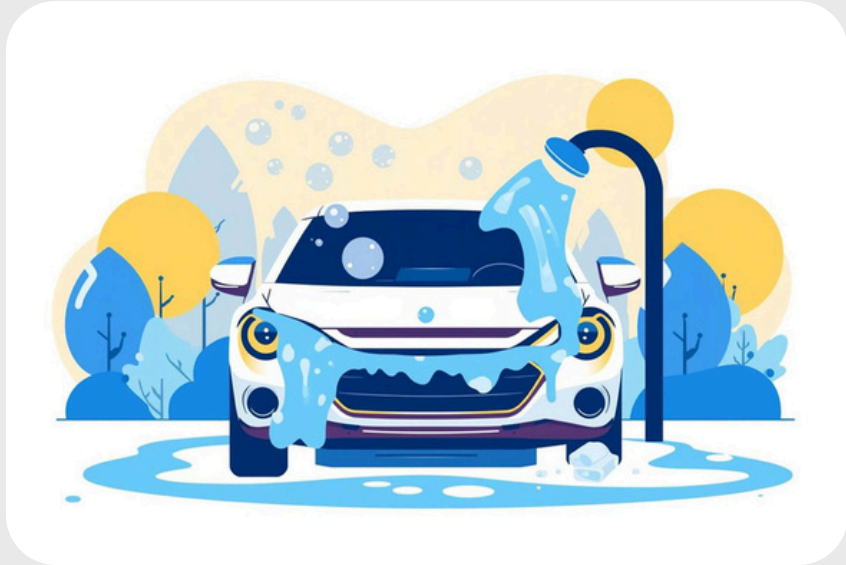


Our Services



Home Cleaning

Enjoy a **sparkling clean** home with our expert services.



Car Washing

Achieve a **shine** that makes your car look new.



Carpentry

Get **custom furniture** built to fit your style.



Electrician

Get **safe electrical** solutions for your home today.



Plumbing

We provide **reliable plumbing** services for any issue.



Appliance Repair

Restore your appliances to **optimal performance** easily.

HARDWARE REQUIREMENTS:

RAM & SSD:8GB, 512 SSD

Processor:i5 or above (or equivalent)





Software Requirements

Frontend: HTML, CSS, JavaScript, ReactJs

Backend: Java (Spring boot 3.3.4), Jakarta persistence API, hibernate

Database: MySQL

IDE's: IntelliJ, Eclipse, Postman, VsCode

Version Control: Git & GitHub

Other Tools: Spring tool suite (STS), maven

Server: Apache tomcat (embedded in spring boot)

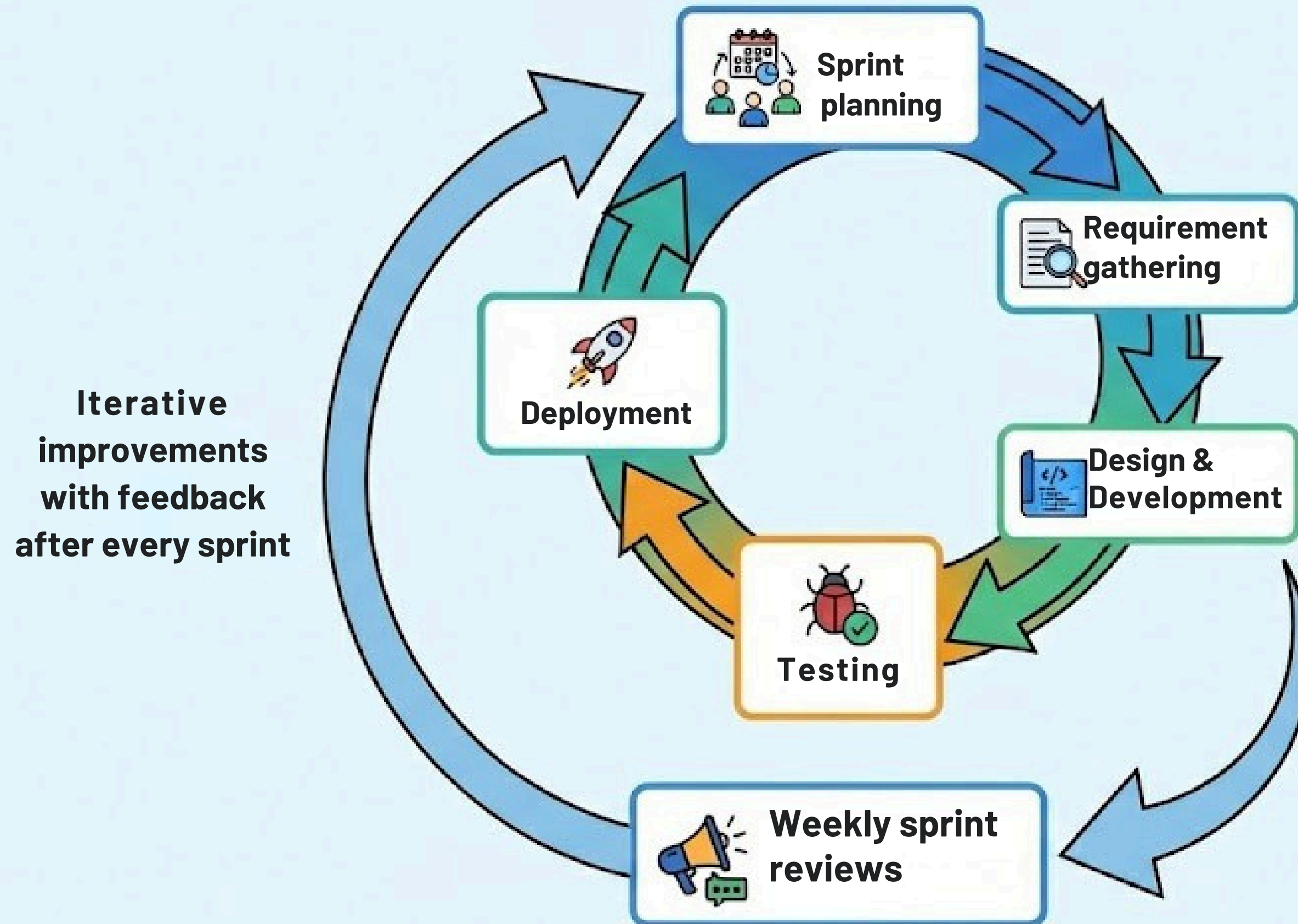
Environment: JDK 21, windows

Browser: chrome, Firefox, or edge

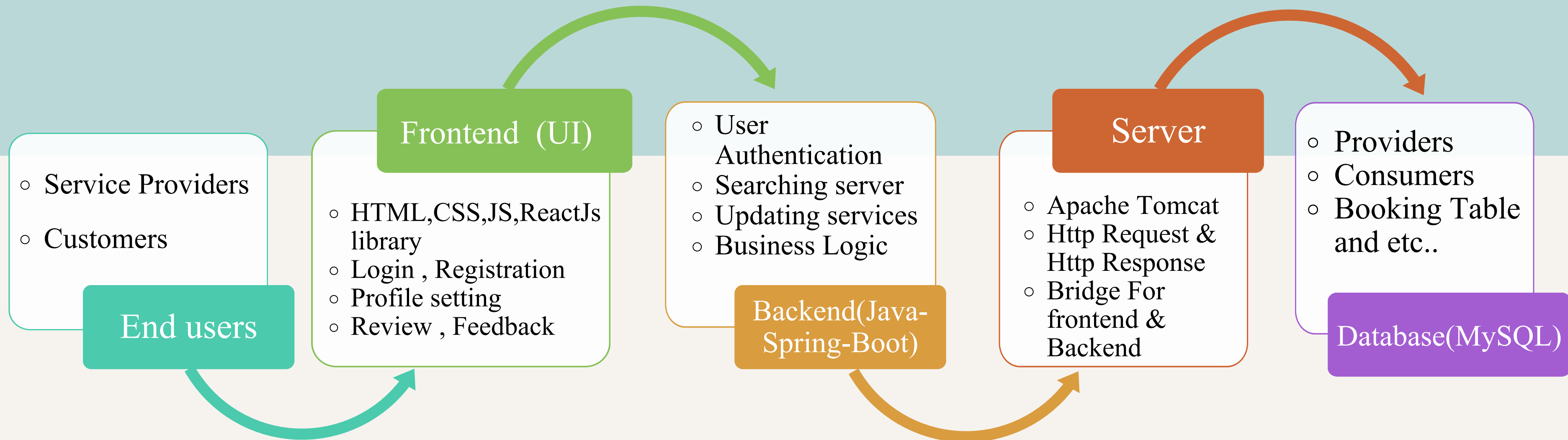
API : Leaflet, Google otp generator,

Methodology

Agile Software Development Approach: Iterative Cycle for Continuous Improvement



System Architecture/Proposed work Block diagram



Functionality of Service Spot

ServiceSpot: Functionality by Role



Map Feature : Connecting Customers with Service Providers



Anya needs a reliable gardener for her new home.

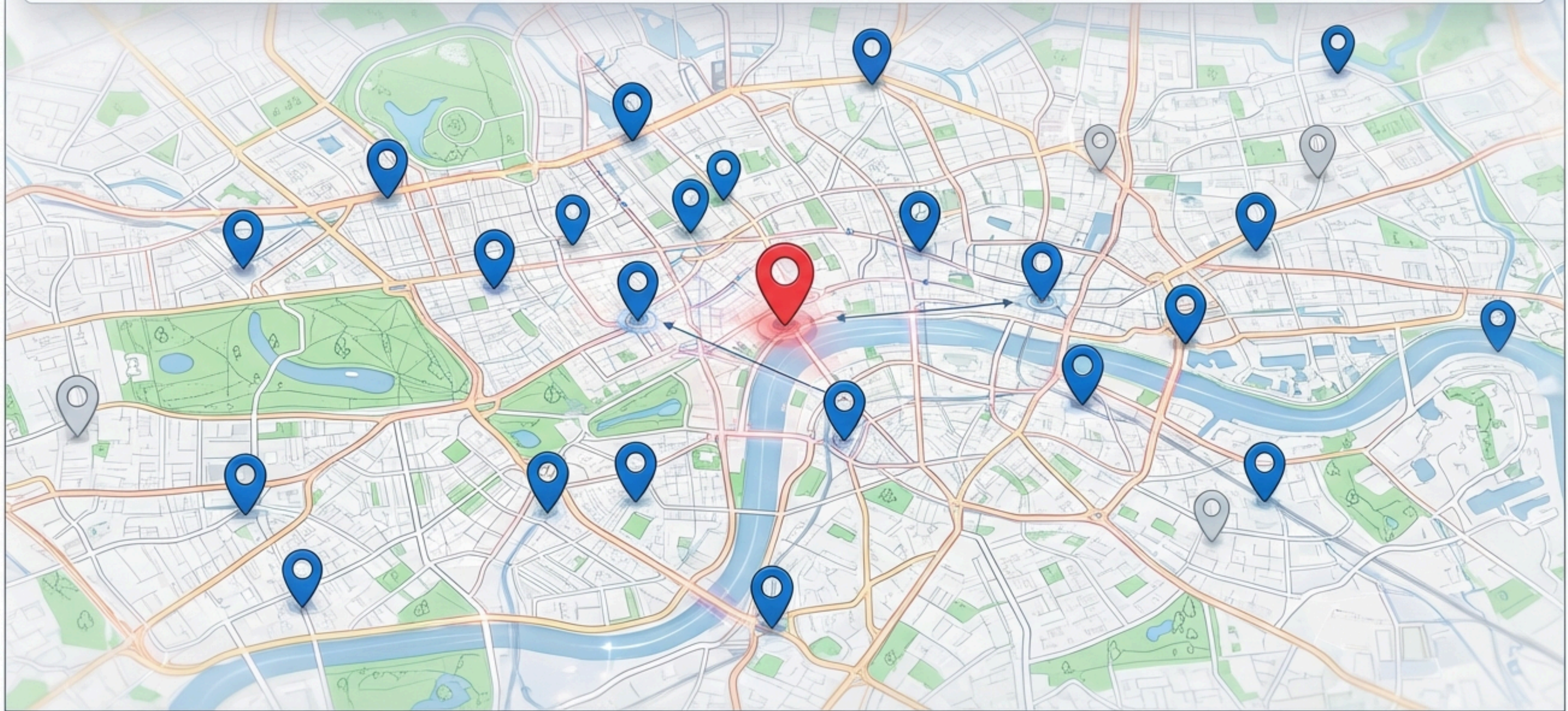
How do we bridge the distance between them, instantly and accurately?



Shilpa runs a local gardening business and wants to find new customers.

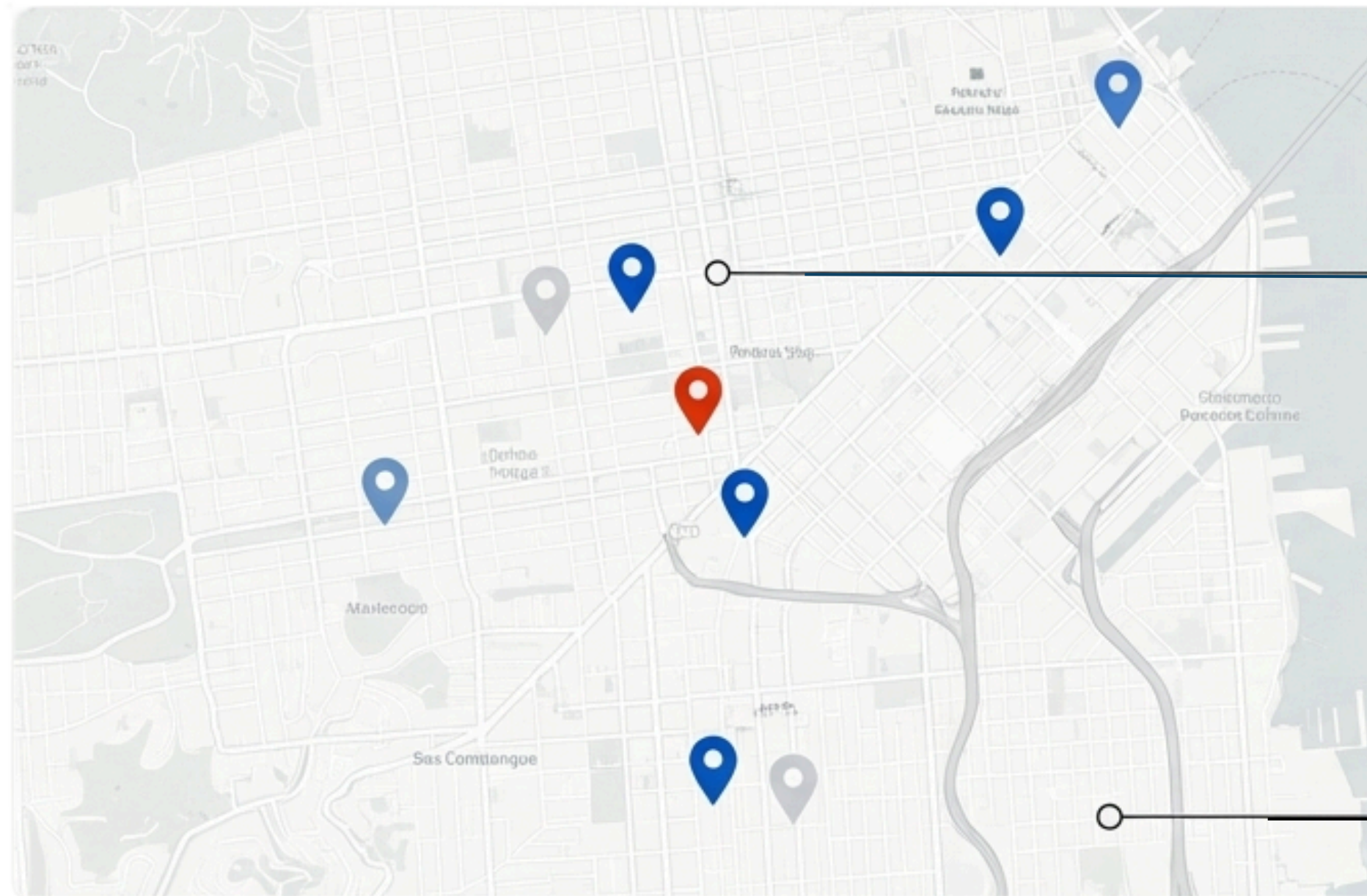
From Search to Solution. Instantly.

Search for providers, services, or locations...



Crafting an Intuitive Visual experience

QuickServe



User's current location
("You are here")



Active providers
(≥ 1 active service)



Inactive providers
(0 active services)

```
<MapContainer center=(position) zoom={13} scrollWheelZoom={true}>  
<TileLayer url="https://{s}.tile.openstreetmap.org/{z}/{x}/{y}.png" />  
</MapContainer>
```

Visual differentiation between active and inactive providers allows users to assess availability at a glance.

Notification feature for Instant, Persistent Communication



Real-Time Delivery

- **Instant Notifications:** WebSocket-based delivery for zero-latency updates.
- **Live Badge Counter:** Real-time unread count updates without page reloads.
- **Dynamic User Detection:** Instantly detects login/logout across tabs for seamless session management.
- **Browser Notifications:** Native browser notification support for engagement outside the app.



Persistent & Reliable

- **Database Storage:** All notifications are permanently stored in MySQL.
- **Track Read/Unread Status:** Granular tracking of user interaction.
- **Complete Notification History:** Users can access all past notifications.
- **Individual & Bulk Actions:** Mark as read or delete notifications one by one or all at once.



Enhanced User Experience

- **Priority Levels:** Classify notifications as HIGH, NORMAL, or LOW.
- **Actionable URLs:** Each notification links directly to the relevant page.
(e.g., a booking or a review).
- **Rich Content:** Integrates review comments, star ratings, and sender information.
- **Categorization:** Users can filter their view by 'all' or 'unread' notifications.

The User's View :



All

Unread

Title: **New Review Received**

Message: Ahmed has left a 5-star review for Plumbing Service.

Relative Timestamp: 2m ago

Title: **System Alert**

Message: Scheduled maintenance in 1 hour.

Relative Timestamp: 10m ago

✓ Mark as Read

🗑 Delete

Title: **Message from Support**

Message: Your ticket #12345 has been updated.

Relative Timestamp: 35m ago

Title: **Booking Confirmation**

Message: Your booking for tomorrow is confirmed.

Relative Timestamp: 1h ago



Admin

Monitors platform health and activity.

- "NEW_CUSTOMER_REGISTERED" (Priority: **NORMAL**)
- "NEW_PROVIDER_REGISTERED" (Priority: **HIGH**)
- "CONTACT_FORM_SUBMITTED" (Priority: **HIGH**)



Service Provider

Manages bookings and service reputation.

- "BOOKING_CREATED" (Priority: **HIGH**)
- "BOOKING_CANCELLED" (Priority: **HIGH**)
- "REVIEW_RECEIVED" (Priority: **NORMAL**)



Customer

Tracks their service requests and history.

- "BOOKING_CONFIRMED" (Priority: **HIGH**)
- "BOOKING_CANCELLED" (Priority: **HIGH**)
- "BOOKING_COMPLETED" (Priority: **NORMAL**)

One System, Two Critical Functions

1. Email Verification

For new Customer' & 'Provider' registrations, ensuring a valid and accessible email address.

2. Password Reset

A secure mechanism for users to regain access to their accounts.



**'6-Digit
Numeric'**

The format of the OTP.



**'10-Minute
Expiry'**

The validity window for
each code.



**'60-Second
Cooldown'**

The rate limit for the
'Resend' functionality.



'SecureRandom'

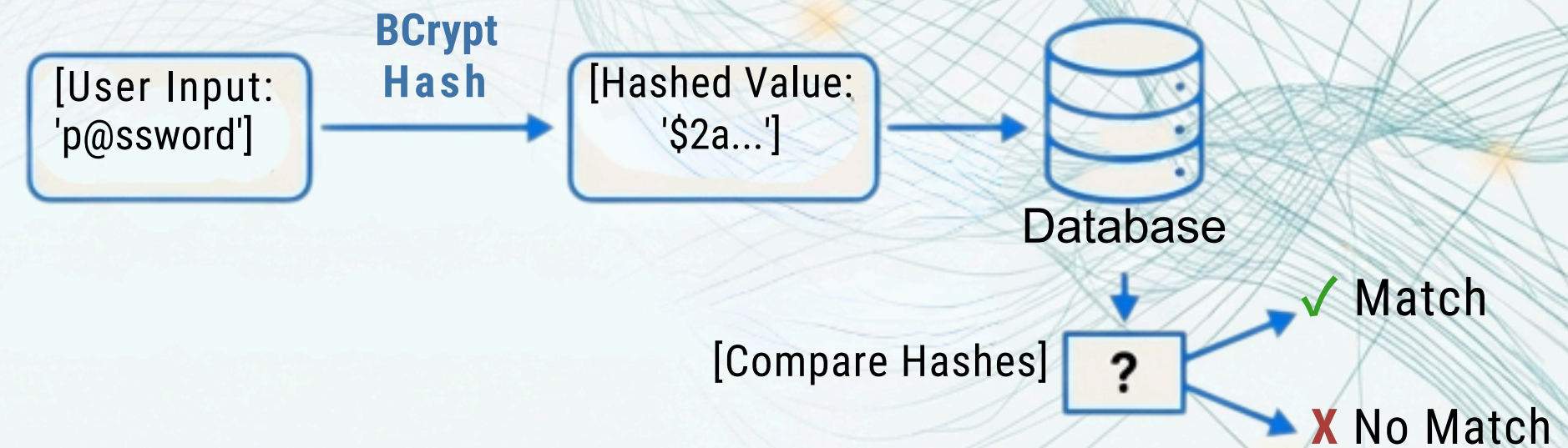
The cryptographic foundation
for OTP generation.

Hardened Data Protection: Why We Use Both Hashing & Encryption

We've implemented two industry standard cryptographic technologies to protect data

One-Way Hashing for Passwords (BCrypt)

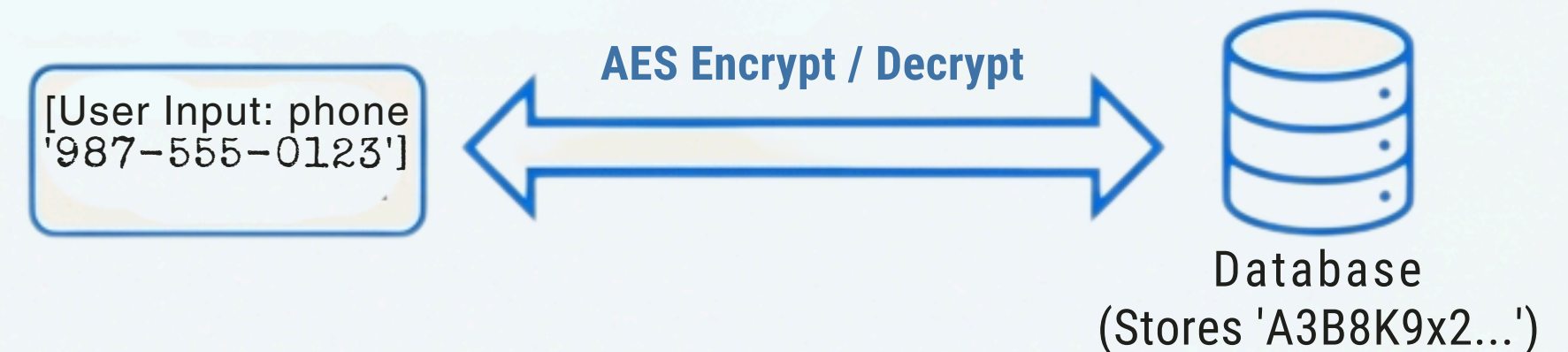
- **The Principle:** We should never be able to retrieve a user's original password.
- **The Goal:** We only need to **VERIFY** if a submitted password matches the stored hash. We don't need to see the password itself.



A one-way process ensures that even if the database is compromised, the actual passwords are not exposed.

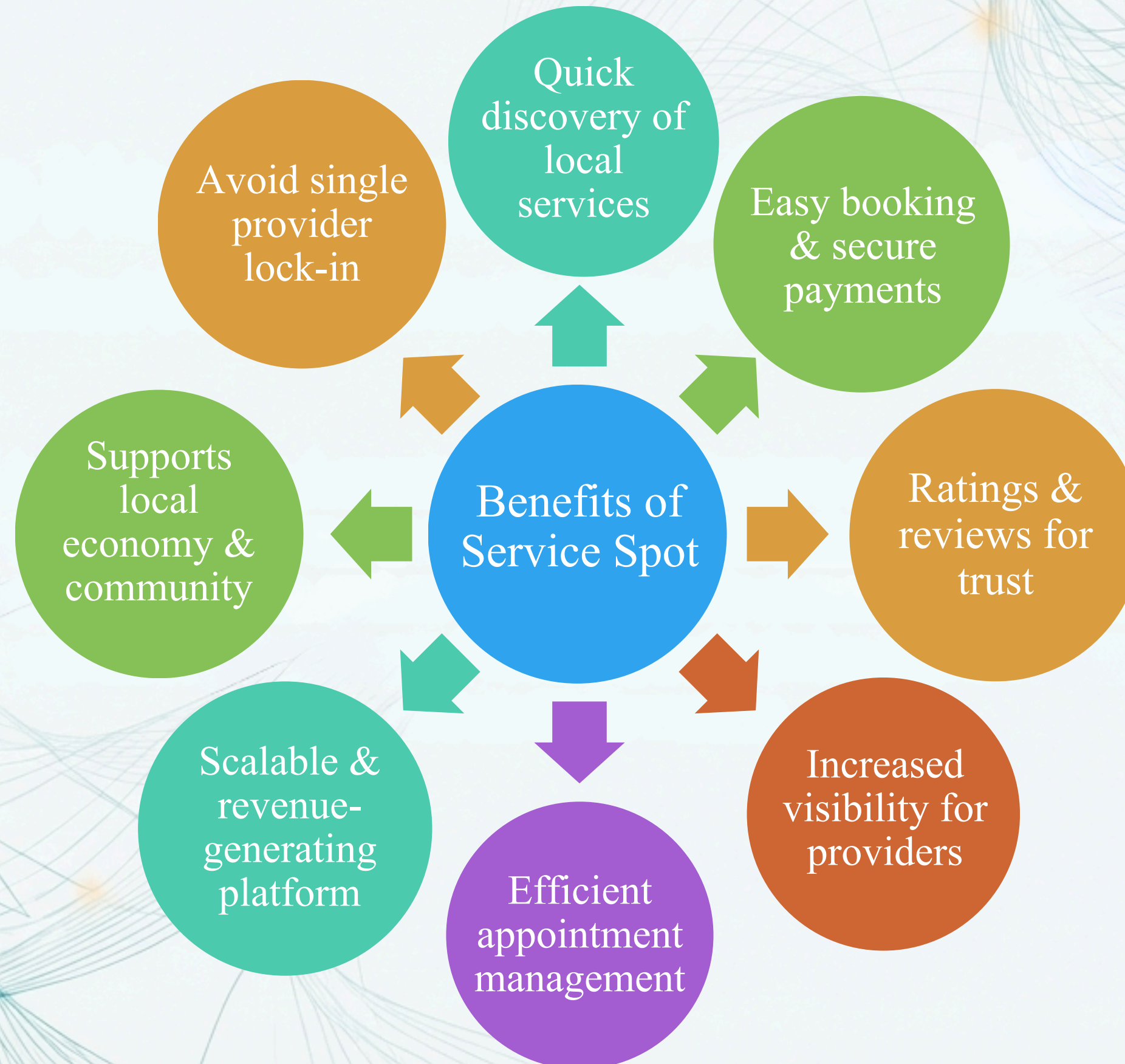
Two-Way Encryption for Phone Numbers (AES)

- **The Principle:** We need to protect this data, but we also need to be able to use it.
- **The Goal:** We need to **RETRIEVE** the original phone number to display it to users or use it for notifications.



This two-way process keeps Personally Identifiable Information (PII) secure at rest.

Benefits of Service Spot



Future Enhancements

1. Payment Integration
2. Provider Enhancements like Portfolio/Gallery.
3. Earnings Dashboard.
4. Live Chat In-app messaging with providers.
5. Mobile App for iOS & Android mobile apps.

Conclusion

The Service Spot localized service discovery and booking app provides a seamless platform connecting users with reliable local service providers. It enhances convenience, trust, and efficiency for customers, while enabling service providers to increase visibility and manage operations effectively and removes single provider lock-in. The scalable and user-friendly solution not only drives business growth but also supports the local economy, fostering a stronger and more connected community. Quick serve exemplifies how technology can simplify daily life while empowering both users and service providers.

The background features a light blue gradient with intricate, flowing teal lines that create a sense of movement and depth. Scattered throughout are small, glowing orange dots, some of which are connected by thin lines, resembling a network or a constellation. The overall aesthetic is modern and digital.

Thank you !!